# Experiment: Measurement of Return Loss

## Objective

To measure the return loss of an RF system by determining the reflected voltage with respect to the incident voltage at different frequencies.

## Components Required

RF signal generator, directional coupler/reflectometer setup, oscilloscope or measuring instrument, RF cables and connectors, and suitable 50 Ω termination/load.

## Theory

Return loss is a measure of the amount of incident RF signal that is reflected back from a load or device because of impedance mismatch. The reflection coefficient is given by:

$$\Gamma = V_{ref} / V_{inc}$$

**Return Loss (RL) = $-20 \log_{10}|\Gamma|$**

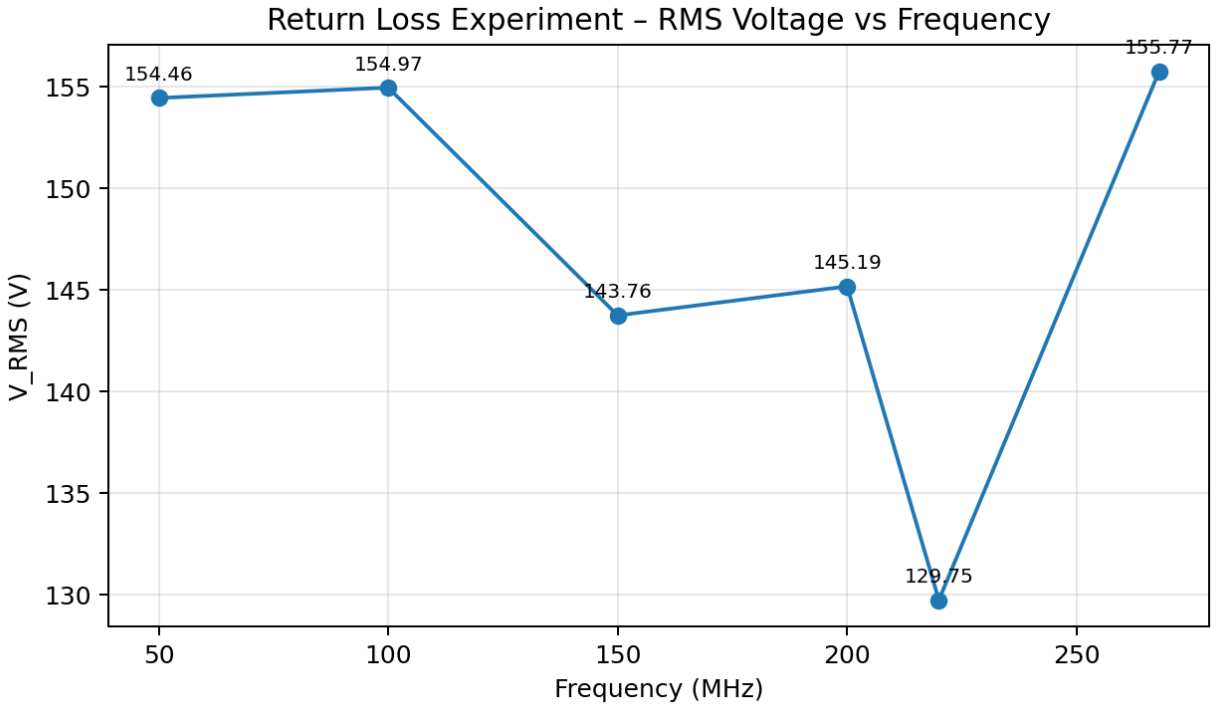
A higher return-loss value indicates better impedance matching and less reflected power.

## Experimental Setup

The RF source was connected to the return-loss measurement setup. The incident voltage was maintained at approximately 1 V. The RMS voltage was recorded at different frequencies. The reflected voltage is required to calculate the reflection coefficient and return loss.

## Observation & Graphs

| Frequency (MHz) | V_RMS (V) | V_inc (V) | V_ref (V) | Return Loss (dB) |
|---|---|---|---|---|
| 50 | 154.46 | 1.00 | — | — |
| 100 | 154.97 | 1.00 | — | — |
| 150 | 143.76 | 1.00 | — | — |
| 200 | 145.19 | 1.00 | — | — |
| 220 | 129.75 | 1.00 | — | — |
| 268 | 155.77 | 1.00 | — | — |


Return Loss Experiment – RMS Voltage vs Frequency
V_RMS (V)
Frequency (MHz)
154.46
154.97
143.76
145.19
129.75
155.77
155
150
145
140
135
130
50
100
150
200
250

## Calculation & Results

**Given:**

$V_{inc} = 1\ V$

**Reflection coefficient:**

$\Gamma = V_{ref} / V_{inc}$

**Return Loss:**

$RL = -20 \log_{10}|\Gamma|$

The recorded RMS-voltage values were 154.46 V, 154.97 V, 143.76 V, 145.19 V, 129.75 V and 155.77 V at 50, 100, 150, 200, 220 and 268 MHz, respectively.

**Note:** The $V_{ref}$ column in the supplied observation sheet is blank. Therefore, numerical return-loss values cannot be calculated reliably from the supplied readings without inventing data.

## Conclusion (Key Point / Takeaway)

The return-loss measurement setup was studied over the given frequency range. RMS voltage readings were recorded at different frequencies. Return loss is determined from the ratio of reflected voltage to incident voltage; hence $V_{ref}$ measurements are required to obtain the final numerical return-loss values.